MATH 128 - HW4 - Jie Wen

Problem 1: Problems from textbook Section 2.1

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$$p_n - p_{n-1} = \frac{1}{n}.$$

As $n \to \infty$,

$$\lim_{n\to\infty} (p_n - p_{n-1}) = \lim_{n\to\infty} \frac{1}{n} = 0.$$

So the difference between consecutive terms tends to 0.

- Group the terms:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots$$

- Estimate each group:

$$\begin{aligned} \frac{1}{3} + \frac{1}{4} &> \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, \\ \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} &> 4 \cdot \frac{1}{8} = \frac{1}{2}, \\ &\vdots \end{aligned}$$

- Each group (after the first two terms) contributes more than $\frac{1}{2}$. Since there are infinitely many such groups, the total sum becomes arbitrarily large:

$$p_n > 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots \to \infty.$$

  Therefore, although

$$\lim_{n\to\infty} (p_n - p_{n-1}) = 0,$$

  the sequence $\{p_n\}$ diverges because the harmonic series diverges.

Problem 2: Problems from textbook Section 2.2

1d

The fixed point condition:

$$\frac{3p^4 + 2p^2 + 3}{4p^3 + 4p - 1} = p.$$

Multiply both sides by the denominator $4p^3 + 4p - 1$:

$$3p^4 + 2p^2 + 3 = p(4p^3 + 4p - 1).$$

Expand the right-hand side:

$$3p^4 + 2p^2 + 3 = 4p^4 + 4p^2 - p.$$

Bring all terms to one side:

$$3p^4 + 2p^2 + 3 - 4p^4 - 4p^2 + p = 0.$$

Simplify:

$$p^4 + 2p^2 - p - 3 = 0.$$

The function

$$f(x) = x^4 + 2x^2 - x - 3$$

satisfies

$$f(p) = p^4 + 2p^2 - p - 3.$$

Thus,

$$f(p) = 0.$$

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Showing the Unique Fixed Point Using **Theorem 2.3**

The function $g(x) = 2^{-x}$ is continuous on $[\frac{1}{3}, 1]$ as it is an exponential function.

Evaluate the range: $g(\frac{1}{3}) = 2^{-1/3} \approx 0.7937$ and $g(1) = 2^{-1} = 0.5$. Since g is decreasing, the image of $[\frac{1}{3}, 1]$ under g is $[0.5, 0.7937]$, which is contained in $[\frac{1}{3}, 1]$.

By part (i) of **Theorem 2.3**, g has at least one fixed point in $[\frac{1}{3}, 1]$.

$g'(x) = -\ln(2) \cdot 2^{-x}$, so $|g'(x)| = \ln(2) \cdot 2^{-x}$.

On $(\frac{1}{3}, 1)$, $|g'(x)|$ is maximized at $x = \frac{1}{3}$: $\ln(2) \cdot 2^{-1/3} \approx 0.5502 < 1$.

Set $k = 0.5502 < 1$. By part (ii) of **Theorem 2.3**, there is exactly one fixed point in $[\frac{1}{3}, 1]$.

Fixed-Point Iteration Approximation Accurate to 10^{-4}

MATLAB codes:

```
%% Fixed-Point Iteration for g(x) = 2^(-x)
% This script finds the unique fixed point of g(x) = 2^(-x) on [1/3, 1]
clear; clc; close all;

% Define the iteration function and parameters
g = @(x) 2.^(-x);          % Iteration function
a = 1/3; b = 1;            % Interval [a, b]
p0 = (a + b)/2;            % Initial guess (midpoint)
tol = 1e-4;                % Tolerance (10^-4)
max_iter = 100;            % Maximum allowed iterations

% Theoretical convergence analysis
k = 2^(-1/3) * log(2);     % Maximum derivative value
fprintf('Theoretical convergence parameter k = %.4f\n', k);

% Estimate required iterations using Corollary 2.5
p1 = g(p0);
error_bound = @(n) (k.^n / (1 - k)) * abs(p1 - p0);
n_theoretical = 0;
while error_bound(n_theoretical) > tol
    n_theoretical = n_theoretical + 1;
end
fprintf('Theoretical iterations required: %d\n', n_theoretical);

% Perform fixed-point iteration
fprintf('\n%-5s %-12s %-12s %-12s\n', 'n', 'p_n', '|p_n - p_{n-1}|', 'Error Bound');
p_prev = p0;
fprintf('%-5d %-12.8f %-12s %-12s\n', 0, p0, 'N/A', 'N/A');

for n = 1:max_iter
    p_current = g(p_prev);
    error_iter = abs(p_current - p_prev);

    % Calculate current error bound
    current_error_bound = error_bound(n);

    fprintf('%-5d %-12.8f %-12.8f %-12.8f\n', n, p_current, error_iter, current_error_bound);

    % Check convergence
    if error_iter < tol
        fprintf('\nConverged after %d iterations!\n', n);
        fprintf('Fixed point approximation: p = %.8f\n', p_current);
        fprintf('Final error: |p_%d - p_%d| = %.8f\n', n, n-1, error_iter);
        break;
    end

    p_prev = p_current;

    if n == max_iter
        fprintf('\nMaximum iterations reached without convergence.\n');
    end
end
```

Output:

```
Theoretical convergence parameter k = 0.5502
Theoretical iterations required: 12

n     p_n          |p_n - p_{n-1}| Error Bound
0     0.66666667   N/A            N/A
1     0.62996052   0.03670615     0.04489049
2     0.64619410   0.01623357     0.02469656
3     0.63896371   0.00723038     0.01358685
4     0.64217406   0.00321035     0.00747482
5     0.64074665   0.00142740     0.00411228
6     0.64138092   0.00063427     0.00226238
7     0.64109901   0.00028192     0.00124465
8     0.64122430   0.00012529     0.00068475
9     0.64116861   0.00005568     0.00037671

Converged after 9 iterations!
Fixed point approximation: p = 0.64116861
Final error: |p_9 - p_8| = 0.00005568
```

Estimate Using **Corollary 2.5** and Comparison

From the MATLAB output:

- **Convergence parameter**: $k = 0.5502$
- **Initial step difference**: $|p_1 - p_0| = 0.03670615$
- **Tolerance**: $\text{tol} = 10^{-4}$

Using the error bound from **Corollary 2.5**:

$$|p_n - p| \leq \frac{k^n}{1-k}|p_1 - p_0|$$

Now need to find the smallest n s.t.

$$\frac{0.5502^n}{1-0.5502} \times 0.03670615 < 10^{-4}$$

$$n\ln(0.5502) < \ln(0.001225)$$

$$n > 11.22$$

Therefore, the **theoretical estimate** suggests that $n \geq 12$ iterations are required.

Metric	Theoretical Estimate	Actual Result
Required Iterations	12	9
Final Error	$< 10^{-4}$ (guaranteed)	5.568×10^{-5}
Convergence Rate	Based on worst-case $k = 0.5502$	Faster

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Part a: $x = \frac{2-e^x+x^2}{3}$

Determining the Interval [a, b]

The problem suggests using the starting interval [0, 1]. Define $g(x) = \frac{2-e^x+x^2}{3}$.

- Evaluate the mapping:
  - $g(0) = \frac{1}{3} \approx 0.333$
  - $g(1) = \frac{3-e}{3} \approx 0.094$
  - Since $g(x)$ is continuous and decreasing (as $g'(x) = \frac{-e^x+2x}{3} < 0$ in [0, 1]), $g([0,1]) \subseteq [0.094, 0.333] \subseteq [0,1]$.
- Check the derivative:
  - $|g'(x)| = \frac{e^x-2x}{3}$.
  - The maximum value in [0, 1] is at $x = 0$, where $|g'(0)| = \frac{1}{3} \approx 0.333 < 1$.

Thus, fixed-point iteration converges to a unique fixed point in [0, 1].

Estimating the Number of Iterations

Use **Corollary 2.5**: $|x_n - p| \leq k^n \max\{x_0 - a, b - x_0\}$, where $k = \max|g'(x)| = \frac{1}{3}$.

Pick $x_0 = 0.5$: now need to solve $(\frac{1}{3})^n \leq 2 \times 10^{-5}$: $n \geq 9.84$

Thus, $n = 10$ iterations are sufficient to guarantee accuracy within 10^{-5}, regardless of the starting point in [0, 1].

Part b: $x = \frac{5}{x^2} + 2$

Determining the Interval [a, b]

Define $g(x) = \frac{5}{x^2} + 2$.

- Evaluate the mapping:
  - $g(2.5) = \frac{5}{6.25} + 2 = 2.8$
  - $g(3) = \frac{5}{9} + 2 \approx 2.556$
  - Since $g(x)$ is continuous and decreasing (as $g'(x) = -\frac{10}{x^3} < 0$), $g([2.5, 3]) \subseteq [2.556, 2.8] \subseteq [2.5, 3]$.
- Check the derivative:
  - $|g'(x)| = \frac{10}{x^3}$.
  - The maximum value in [2.5, 3] is at $x = 2.5$, where $|g'(2.5)| = \frac{10}{15.625} = 0.64 < 1$.

Thus, fixed-point iteration converges to a unique fixed point in [2.5, 3].

Estimating the Number of Iterations

Use **Corollary 2.5**: $|x_n - p| \leq k^n \max\{x_0 - a, b - x_0\}$, where $k = 0.64$.

Pick $x_0 = 2.75$: now need to solve $0.64^n \leq 4 \times 10^{-5}$. Thus, $n \geq 22.69$ ($n = 23$) iterations are sufficient to guarantee accuracy within 10^{-5}, regardless of the starting point in [2.5, 3].

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Consider the function:

$$g(x) = x^2 + \frac{1}{4}$$

Check for Fixed Point

Solve:

$$x = x^2 + \frac{1}{4}$$

Rewriting:

$$x^2 - x + \frac{1}{4} = 0$$

This factors as:

$$(x - \tfrac{1}{2})^2 = 0$$

So the only solution is:

$$x = \frac{1}{2}$$

Thus, g has a **unique fixed point** at $x = \frac{1}{2}$.

Verify Failure of Hypotheses

Hypothesis (i)

- $g(x) = x^2 + \frac{1}{4}$ is continuous on $[0, 1]$.
- But:

$$g(1) = 1^2 + \frac{1}{4} = \frac{5}{4} > 1$$

So $g(x) \notin [0, 1]$ for all $x \in [0, 1]$, hypothesis (i) fails.

Hypothesis (ii)

- Compute the derivative:

$$g'(x) = 2x$$

- On $(0, 1)$, $|g'(x)| = 2x$, which approaches 2 as $x \to 1^-$.
- No constant $k < 1$ can bound $|g'(x)|$ on $(0, 1)$, hypothesis (ii) fails.

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(a) Theorem 2.3 is true

Assume, for contradiction, that g has **two distinct fixed points** p and q in $[a, b]$, with $p < q$. Then:

$$g(p) = p \quad \text{and} \quad g(q) = q.$$

Since g is differentiable on (a, b), by the **Mean Value Theorem**, there exists some $c \in (p, q) \subset (a, b)$ such that:

$$g'(c) = \frac{g(q) - g(p)}{q - p} = \frac{q - p}{q - p} = 1.$$

But the modified condition requires:

$$g'(x) \leq k < 1 \quad \text{for all } x \in (a, b).$$

In particular, $g'(c) \leq k < 1$, which contradicts $g'(c) = 1$.

Contradiction! Hence, our assumption that there are two distinct fixed points is false.

Therefore, under the condition $g'(x) \leq k < 1$ for all $x \in (a, b)$, the fixed point of g in $[a, b]$ is **unique**.

(b) Theorem 2.4 may not hold

Consider the function

$$g(x) = 1 - x^2 \quad \text{on } [0, 1].$$

- Check the conditions:
  - **Continuity and range**:
    g is continuous on $[0, 1]$.
    For $x \in [0, 1]$,

    $$g(x) = 1 - x^2 \in [0, 1],$$

    so $g(x) \in [0, 1]$ for all $x \in [0, 1]$.
  - **Derivative condition**:

    $$g'(x) = -2x.$$

    On $(0, 1)$: $g'(x) \leq 0$.
    Choose $k = 0.5$. Then

    $$g'(x) \leq 0 \leq 0.5 = k,$$

    so the modified condition $g'(x) \leq k$ is satisfied.

The fixed points satisfy $x = g(x)$, i.e.,

$$x = 1 - x^2 \quad \Rightarrow \quad x^2 + x - 1 = 0.$$

Solving gives

$$x = \frac{-1 \pm \sqrt{5}}{2}.$$

The positive root is

$$p = \frac{\sqrt{5} - 1}{2} \approx 0.618.$$

Now consider the iteration with $p_0 = 0$:

$$\begin{aligned} p_1 &= g(0) = 1, \\ p_2 &= g(1) = 0, \\ p_3 &= g(0) = 1, \quad \text{and so on.} \end{aligned}$$

The sequence oscillates between 0 and 1 and does not converge to the fixed point p and fails to converge.

Problem 3: Consistent fixed point formulations

MATLAB codes:

```
%% Fixed-Point Iteration Analysis for Three Functions
% This code tests the convergence of different fixed-point formulations.

% Known exact solution
x_exact = 0.56714329040978;

% Define the initial guess
x0 = 0.5;

% Define the three G(x) functions
G1 = @(x) exp(-x);
G2 = @(x) (1 + x) ./ (1 + exp(x));
G3 = @(x) x + 1 - x .* exp(x);

% Initialize arrays to store results for each method
num_iterations = 10;
x1 = zeros(1, num_iterations + 1);
x2 = zeros(1, num_iterations + 1);
x3 = zeros(1, num_iterations + 1);
error1 = zeros(1, num_iterations + 1);
error2 = zeros(1, num_iterations + 1);
error3 = zeros(1, num_iterations + 1);

% Set the initial values
x1(1) = x0;
x2(1) = x0;
x3(1) = x0;
error1(1) = abs(x0 - x_exact);
error2(1) = abs(x0 - x_exact);
error3(1) = abs(x0 - x_exact);

% Perform fixed-point iterations
for k = 1:num_iterations
    % Iteration for G1
    x1(k+1) = G1(x1(k));
    error1(k+1) = abs(x1(k+1) - x_exact);

    % Iteration for G2
    x2(k+1) = G2(x2(k));
    error2(k+1) = abs(x2(k+1) - x_exact);

    % Iteration for G3
    x3(k+1) = G3(x3(k));
    error3(k+1) = abs(x3(k+1) - x_exact);
end

% Display the results in a table
iteration_num = (0:num_iterations)';
T = table(iteration_num, x1', error1', x2', error2', x3', error3', ...
    'VariableNames', {'Iteration', 'x_G1', 'Error_G1', 'x_G2', 'Error_G2', 'x_G3', 'Error_G3'});
disp(T);

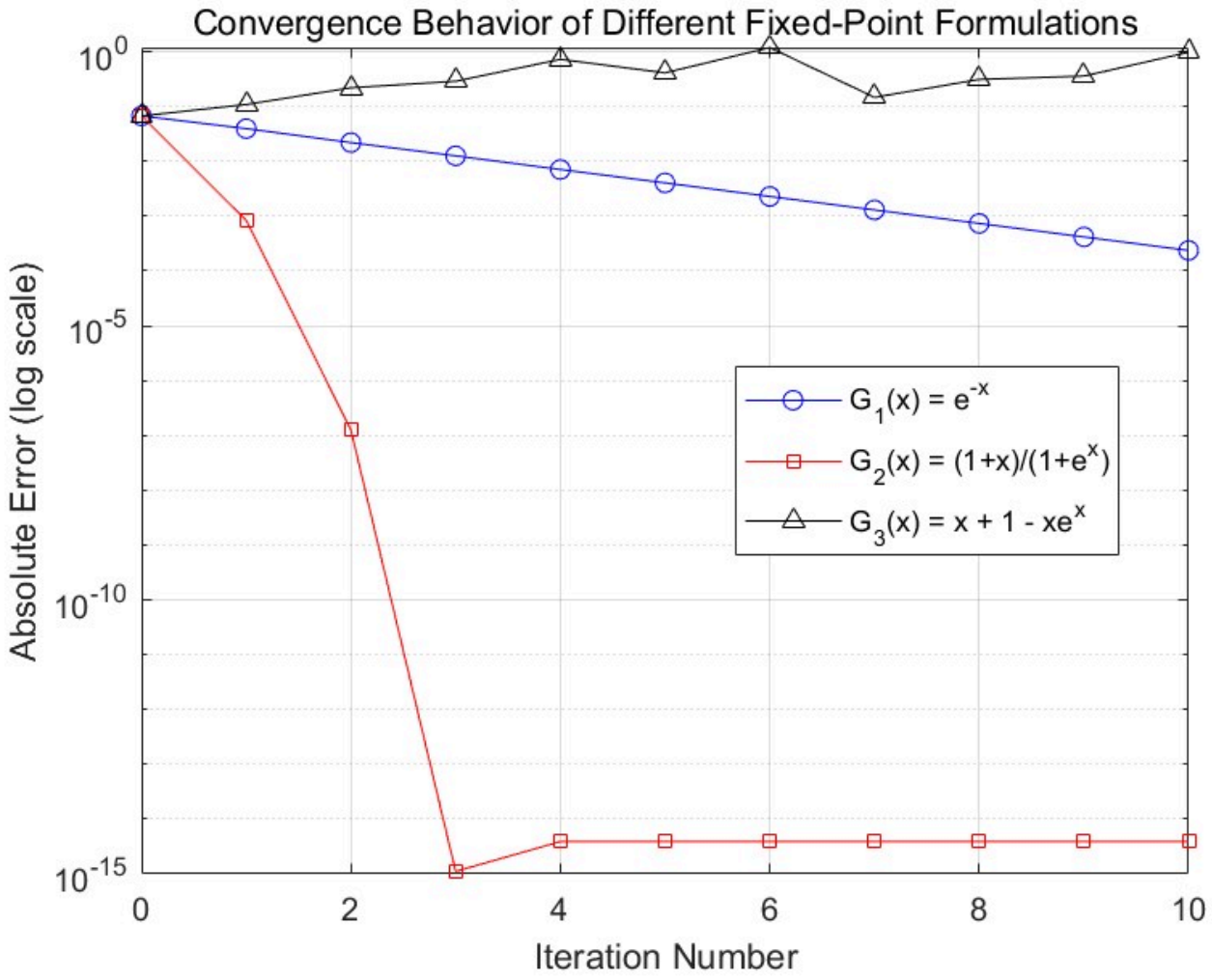
% Plot the error convergence on a semi-log plot to see the trend clearly
figure;
semilogy(iteration_num, error1, 'b-o', 'DisplayName', 'G_1(x) = e^{-x}');
hold on;
semilogy(iteration_num, error2, 'r-s', 'DisplayName', 'G_2(x) = (1+x)/(1+e^x)');
semilogy(iteration_num, error3, 'k-^', 'DisplayName', 'G_3(x) = x + 1 - xe^x');
hold off;
grid on;
xlabel('Iteration Number');
ylabel('Absolute Error (log scale)');
title('Convergence Behavior of Different Fixed-Point Formulations');
legend('Location', 'best');
```

Output:

```
Iteration     x_G1       Error_G1      x_G2       Error_G2       x_G3      Error_G3
_________    _______    __________    _______    __________    ________    ________

    0           0.5       0.067143       0.5       0.067143         0.5    0.067143
    1       0.60653       0.039387    0.56631    0.00083229     0.67564      0.1085
```

```
 2        0.54524      0.021904     0.56714     1.2537e-07     0.34781      0.21933
 3         0.5797       0.01256     0.56714     1.1102e-15     0.85532      0.28818
 4        0.56006     0.0070787     0.56714     3.8858e-15    -0.15651      0.72365
 5        0.57117     0.0040289     0.56714     3.8858e-15     0.97733      0.41018
 6        0.56486     0.0022803     0.56714     3.8858e-15    -0.61976       1.1869
 7        0.56844     0.0012948     0.56714     3.8858e-15     0.71371      0.14657
 8        0.56641    0.00073384     0.56714     3.8858e-15     0.25663      0.31052
 9        0.56756    0.00041634     0.56714     3.8858e-15     0.92492      0.35778
10        0.56691    0.00023608     0.56714     3.8858e-15    -0.40742      0.97457
```



Based on the output: analyze the convergence of the three fixed-point formulations as follows.

Formulation	Convergence Status	Convergence Rate	Key Observation
$G_1(x) = e^{-x}$	Converges	**Linear**	Error decreases steadily (~0.0013 after 10 iterations)
$G_2(x) = \frac{1+x}{1+e^x}$	Converges	**Extremely Fast (Quadratic)**	Reaches ~(10^{-15}) error after just 2 iterations
$G_3(x) = x + 1 - xe^x$	**Diverges**	N/A	Iterates oscillate wildly; error increases

The convergence behavior is determined by the magnitude of the derivative $|G'(x^*)|$ at the fixed point x^*.

- **Why G_1 Converges Slowly**: The derivative of G_1 is $G_1'(x) = -e^{-x}$. At the fixed point $x^* \approx 0.567$, the magnitude is $|G_1'(x^*)| \approx 0.567$. Since this is less than 1, the method converges, but the rate is linear because the derivative is not zero.
- **Why G_2 Converges Very Quickly**: The derivative $G_2'(x^*)$ is very close to zero. When $|G'(x^*)| = 0$, the method can achieve **quadratic convergence**, meaning the error squares in each iteration. This explains why the error drops to machine precision so rapidly.
- **Why G_3 Diverges**: The derivative of G_3 is $G_3'(x) = 1 - e^x - xe^x$. At x^*, the magnitude $|G_3'(x^*)|$ is greater than 1. This violates the convergence condition for fixed-point iterations, causing small errors to be amplified each step and the iterations to diverge.